

4.1: The Language of Functions:

A mapping transforms one variable to another
Types of mappings:

THIS IS A BAD DEFINITION!

One to One: Each input is mapped to a unique output. $\exists x_1, x_2 \in Dom(f) \mid f(x_1) = f(x_2) \iff x_1 = x_2$
Many to One: Many different inputs can lead to a single outcome
One to Many: A single input is mapped to multiple outputs.
Many to Many: A mapping is both one to many and many to one.

An object is the input to a mapping, and an image is the output.
Only One to One and Many to One mappings are considered functions. Hence only One to One functions have a valid inverse (with no $Dom(f^{-1})$ restrictions).
In a function, there is only one image for one object.

Sets:

$\mathbb{N}$: 0, 1, 2, 3, ...
$\mathbb{Z}$: ..., -2, -1, 0, 1, ...
$\mathbb{Q}$: $\frac{1}{2}, \frac{-3}{2}, \frac{2}{5}, \ldots$
$\mathbb{R}$: π, 1, -1, $\sqrt{2}$

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$
$\mathbb{R} \setminus \mathbb{Q}$ = Set of irrationals.

Domains & Ranges:

The Domain of a function is the set of all possible inputs (inputs).
The Range of a function is the set of all possible images (outputs).

Function Notation:

$f: x \to y \quad f(x) = y$

Transformations:

$f(x) \longmapsto f(x-a)+b \Rightarrow$ Translation $\begin{pmatrix} a \\ b \end{pmatrix}$
$f(x) \mapsto af(x) \Rightarrow$ Stretch a parallel to $x=0$
$f(x) \mapsto f(ax) \Rightarrow$ Stretch $\frac{1}{a}$ parallel to $y=0$

When multiple transformations are done in succession, order matters. For x, always transform first, then stretch.

4.2: Composite & Inverse Functions:

A composite function describes the image of one function acting as the object for another.
If it is the same function, use $f^n(x)$
In order to apply a function repeatedly, it's range must be contained fully within it's domain

For $gf(x)$: $Dom(f) \mapsto Ran(f) \subseteq Dom(g) \mapsto Ran(gf)$
Order of application of function matters.

Inverse Functions:

If One to One, a function can have a fully valid inverse function such that:

$Ran(f) = Dom(f^{-1}) \iff Ran(f^{-1}) = Dom(f)$

Inverse functions are a reflection in line $y=x$, as images are mapped to objects
Hence to find an inverse, rearrange $f(y) = x$, hence $y = f^{-1}(x)$

A many to one function may have an inverse function if the domain of $f(x)$ is restricted to create a one-to-one mapping.

$ff^{-1}(x) = f^{-1}f(x) = x$

$f^{-1}(x) = f(x) \iff (a,b) = (b,a) \Rightarrow a = b$
$\Rightarrow$ Only intersect on line $y=x$

4.3: The Modulus Function:

$$|x| = \begin{cases} x : x \geq 0 \\ -x : x < 0 \end{cases}$$

To sketch the modulus function, if $y<0$, $-y$ is plotted:

$|x| = k \Rightarrow x = \pm k$
$|x| < k \Rightarrow -k < x < k$
$|x| > k \Rightarrow \{x < -k\} \cup \{k < x\}$
Find critical regions, and test values!